

Nick

Software Developer - Test automation · Embedded systems · DevOps

felixfalling.github.io/FelixFallingCoolSite · github.com/FelixFalling

ABOUT

I build software with a tester's mindset - automation, validation, and documentation-driven development, from embedded firmware up to web interfaces.

SELECTED PROJECTS

Rubber Duckie Terminator

TEAM OF 3

Real-time perception-to-actuation pipeline: YOLOv8 trained entirely on synthetic Blender renders (1,000+ auto-labeled images, zero real photos), RealSense depth for metric 3D localization, and an Arduino two-servo laser turret - 85–91% real-world detection.

github.com/FelixFalling/Rubber-Duckie-Terminator · [demo video on the site](#)

Classic AI, From Scratch

COLLECTION

Heuristic search, a genetic 8-queens solver with population-size experiments, a tic-tac-toe game agent, hand-rolled gradient descent, and a transfer-learning CNN (TensorFlow, 97.35%).

github.com/FelixFalling - multiple repositories

Portfolio Site

SOLO

Next.js + TypeScript static site with an animated coastal scene; Playwright test suite (desktop, mobile, and axe-core accessibility scans) run by GitHub Actions CI on every push.

github.com/FelixFalling/FelixFallingCoolSite

SKILLS

Languages: Python, C/C++, TypeScript/JavaScript, Bash

Testing: Test automation, black-box & system validation, Playwright

Embedded: Arduino, Raspberry Pi, serial protocols, OpenCV

DevOps: Docker, Terraform, Ansible, Proxmox, CI/CD, GitHub Actions

Security: Malware analysis, YARA, Suricata, packet analysis

Web & UI: Next.js, React, Node.js, Qt/QML (PySide6)

This is a pseudonymous portfolio résumé. A full résumé - with work history, education, and contact details - is available on request via [GitHub](#).